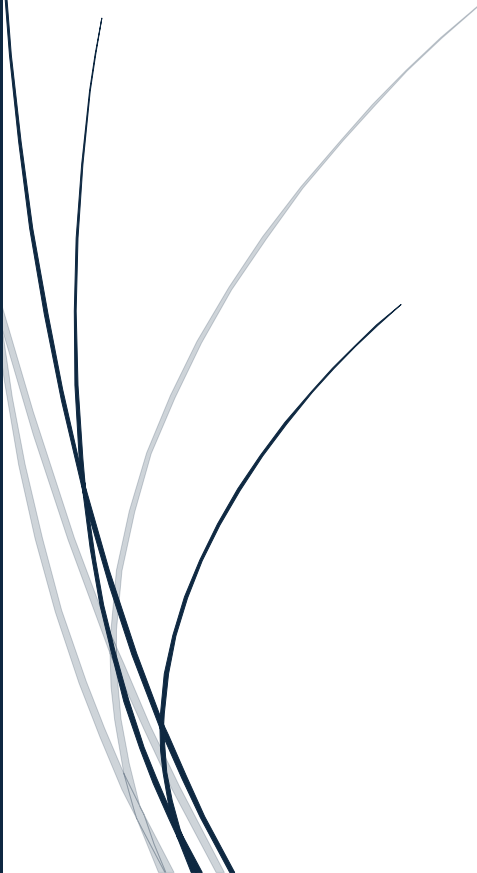




Version 1

Transitioning to the Cloud

Proposal



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ECOGOODS

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1. Introduction

EcoGoods has outgrown its on-premises infrastructure. The business now serves customers across North America, Europe and parts of Asia while relying on one physical server to host its e-commerce platform, custom web application, Dockerised services and MySQL database. This concentrates sales, inventory, orders and customer data on a single system. Traffic spikes from marketing campaigns have already caused slow loading and downtime.

EcoGoods' monthly solid-state drive (SSD) backup practice is inadequate because selected data is copied manually and stored in the same server room, leaving both live systems and backups exposed to shared risks such as hardware failure, ransomware, theft or site damage. Cloud adoption is therefore a business continuity and growth requirement, not simply a technical upgrade. This report proposes a mixed cloud-service architecture and phased migration strategy to improve scalability, availability, recovery, security and global performance while reducing dependence on fragile local infrastructure.

2. Cloud Services Proposal and Evaluation

2.1 Overview of Cloud Service Models

EcoGoods' challenges cannot be addressed by a single cloud service model. The scenario includes a customer-facing e-commerce platform, Dockerised services, a business-critical MySQL database, on-premises email, file storage, HR/finance software, traffic spikes and weak manual backups. The NIST framework categorises cloud services into Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS), helping assess how much management responsibility remains with EcoGoods and how much transfers to the provider (Simmon, 2018).

IaaS provides virtualised compute, storage and networking while the customer manages operating systems, middleware, applications and configuration (Younis *et al.*, 2025). For EcoGoods, IaaS can support temporary migration hosts, testing environments or compatibility workloads. However, it should not be the primary model because patching, database administration, backup scheduling and capacity planning would remain with the business.

PaaS and managed services are more appropriate for the core e-commerce platform because they reduce infrastructure management while supporting custom applications (Younis *et al.*, 2025). Since EcoGoods already uses Docker, Amazon Elastic Kubernetes Service (Amazon EKS) can preserve the existing deployment model while adding orchestration, scaling and resilience. Amazon Relational Database Service (Amazon RDS) can also retain MySQL compatibility while improving backup, failover and recovery.

SaaS is most suitable for non-core functions such as email, office productivity, HR and finance (Younis *et al.*, 2025). These workloads are necessary but do not differentiate EcoGoods as an online retailer. Therefore, EcoGoods should adopt a public-cloud, mixed-service architecture: managed services for the core application and database, storage services for files and backups, and SaaS for internal business functions.

2.2 Alignment with EcoGoods Needs

For compute and application hosting, EcoGoods should run its Dockerised e-commerce application on Amazon EKS, supported by an Application Load Balancer and autoscaling (Mustyala and Allam, 2022). This addresses the single-server capacity problem that causes slow loading and downtime during campaigns. Load balancing distributes requests across healthy targets, while autoscaling adds or removes compute capacity according to demand (Mustyala and Allam, 2022). This addresses the real weakness: lack of elasticity and fault tolerance.

For the database layer, EcoGoods should migrate MySQL to Amazon RDS for MySQL using Multi-Availability Zone (Multi-AZ) deployment (Thodupunuri, 2022). The database stores inventory, orders and customer data, making it business-critical. Multi-AZ deployment maintains database redundancy across Availability Zones and supports automated failover, improving availability compared with a single self-managed database (AWS, 2026h). This also reduces reliance on one IT Manager manually copying selected data to SSD once per month.

For files, static assets and backups, EcoGoods should use Amazon Simple Storage Service (Amazon S3). Product images, exported reports and backup archives are better suited to scalable object storage than local disk storage. S3 is designed for high durability and stores

data redundantly across multiple Availability Zones by default (AWS, 2026i). Lifecycle policies can move older backups to lower-cost storage classes without weakening resilience. For global performance, Amazon CloudFront should act as a content delivery network (CDN). EcoGoods serves customers across North America, Europe and Asia, so users should not depend on content served from one office location. CDNs use distributed cache servers to deliver content closer to users, reducing latency and origin load (Ali, Fang and Khan, 2025; AWS, 2026l). SaaS should also replace on-premises email and, where feasible, desktop HR and finance tools because mature SaaS platforms already provide managed availability, updates and collaboration features (Althani, 2025).

2.3 Proposed Architecture

The proposed architecture should be deployed inside an Amazon Virtual Private Cloud (Amazon VPC). Public subnets should contain only internet-facing components such as the Application Load Balancer and AWS Web Application Firewall (AWS WAF), while private application subnets host EKS workloads and private database subnets contain Amazon RDS. This prevents direct public access to the database and supports defence-in-depth (Alouffi *et al.*, 2021; Thodupunuri, 2022). The workload-to-service mapping is summarised in Appendix Item A, and the proposed architecture is illustrated in Appendix Item B.

The web tier should use Route 53, CloudFront, AWS WAF and an Application Load Balancer. The application tier should run Docker containers in EKS with rolling deployments (Mustyala and Allam, 2022). Secrets should be stored in AWS Secrets Manager rather than plain-text configuration files. The database tier should use RDS for MySQL with Multi-AZ failover, AWS Key Management Service (AWS KMS) encryption and restrictive security groups (Thodupunuri, 2022). CloudWatch, CloudTrail and a continuous integration and continuous delivery (CI/CD) pipeline should support monitoring, auditability, testing and controlled deployment.

2.4 Justification for Combination of Cloud Services

The mixed cloud model is justified because each service addresses a specific weakness in the current environment. Scalability is improved through load balancing, autoscaling, EKS and CloudFront, allowing the platform to respond to campaign-driven traffic spikes without purchasing permanent peak-capacity hardware. Resilience is improved through Multi-AZ RDS, S3 durability, distributed application workloads and automated backups, replacing the single-server and monthly SSD backup approach. These choices align with the AWS Well-Architected reliability guidance, which emphasises resilient architecture, automation and proven failure recovery (AWS, 2026j).

Operational efficiency is improved because managed services reduce the workload on EcoGoods' small IT function (Simmon, 2018). RDS reduces database administration, EKS reduces container orchestration effort, S3 reduces storage maintenance and SaaS removes the need to operate email and office infrastructure. Cost control is also improved because capacity can scale with demand, although budgets, tagging, alerts and regular reviews must be used to prevent cost overruns (Maniah *et al.*, 2022).

Security remains a core design requirement. AWS uses a shared responsibility model in which AWS secures the underlying cloud infrastructure while the customer remains responsible for identity, access, data, configuration and application security (AWS, 2026k). Therefore, Identity and Access Management (IAM) least privilege, multi-factor authentication (MFA), private subnets, security groups, WAF, KMS encryption, monitoring and backup testing must be implemented from the beginning. This is important because cloud migration risk includes information security risk, loss of data access, compliance risk, financial risk and management failure (Maniah *et al.*, 2022). Data leakage and data tampering are also prominent cloud security concerns, reinforcing the need for layered protection (Alouffi *et al.*, 2021). The key security and governance controls supporting this design are summarised in Appendix Item C. Overall, the proposed mixed AWS public-cloud model is more appropriate than a single IaaS migration because it aligns each workload with the service model that best supports scalability, robustness, security and future growth.

3. Cloud Provider Analysis

3.1 Comparison of Cloud Providers

The three most suitable hyperscale providers for EcoGoods are Amazon Web Services (AWS), Microsoft Azure and Google Cloud. All three can support scalable web hosting, managed databases, object storage, identity management and global content delivery; therefore, provider selection should be based on fit with the scenario rather than brand preference alone.

Table 1: Scenario-Based Comparison of Cloud Providers for EcoGoods

Criterion	AWS	Microsoft Azure	Google Cloud
Global infrastructure	39 regions, 123 Availability Zones and a large edge network	70+ regions and 400+ data centres	43 regions and 130 zones
Container service	Amazon EKS	Azure Kubernetes Service	Google Kubernetes Engine
Managed MySQL	Amazon RDS for MySQL	Azure Database for MySQL	Cloud SQL for MySQL
Object storage	Amazon S3	Azure Blob Storage	Google Cloud Storage
Best fit	E-commerce, containers, mature managed services, CDN and resilience	Microsoft enterprise integration and Microsoft 365 identity ecosystem	Kubernetes, analytics and data services

Source: AWS (2026b); Google Cloud (2026); Microsoft (2026a).

AWS is the strongest match because its services map directly to the proposed design: EKS for Docker containers, RDS for MySQL, S3 for storage and backups, CloudFront for global performance, Route 53 for DNS, and IAM/KMS/WAF for security. Azure is strong where Microsoft 365 and Entra ID dominate, while Google Cloud is particularly strong in Kubernetes and analytics. However, EcoGoods' immediate challenge is scaling a Dockerised e-commerce platform and MySQL database, making AWS the most balanced first-choice ecosystem.

3.2 Selected Provider: Amazon Web Services

AWS is recommended because it addresses EcoGoods’ main weaknesses: single-server dependency, traffic spikes, weak backup practice, global latency and limited administration capacity. EKS preserves Docker while adding managed orchestration (AWS, 2026f). RDS for MySQL improves database availability through Multi-AZ failover (Thodupunuri, 2022). S3 and AWS Backup replace monthly SSD backup with automated, off-site protection, while CloudFront, Route 53 and load balancing improve global performance and availability (Ali, Fang and Khan, 2025; AWS, 2026j, 2026l).

The decision is not risk-free. AWS-specific services may create vendor lock-in, and usage-based pricing can become expensive if data transfer, logs, storage or idle resources are poorly governed (Maniah *et al.*, 2022). This should be mitigated through portable design choices, including Docker containers, Kubernetes manifests, MySQL compatibility and exportable backups. Cost governance should also be enforced through budgets, tagging, monitoring and regular architecture reviews (AWS, 2026c). This balanced approach addresses service reliability, global infrastructure, cost and wider operational implications rather than treating cloud as automatically superior.

3.3 Indicative Cost Estimation

The following estimate is indicative only and should be validated using the AWS Pricing Calculator before implementation, because final cost depends on region, traffic volume, storage growth and support requirements.

Table 2: Indicative Monthly Cost Estimate for the Proposed AWS Solution

Service	Assumption	Indicative monthly cost (USD)
EC2/EKS worker nodes	Two small/medium Linux nodes	\$ 60
Amazon EKS	One cluster control plane	\$ 73
RDS MySQL Multi-AZ	Small production database	\$ 150
Amazon S3	500 GB files/backups	\$ 12-15
CloudFront	Moderate traffic/caching	\$ 20

Route 53	DNS hosted zone and queries	\$ 1-3
WAF, monitoring, backup overhead	Basic usage	\$ 30-50
AWS Support	Developer support plan	\$ 29
Estimated monthly total		\$ 375-400

Source: AWS Pricing Calculator assumptions (AWS, 2026c).

This is more expensive than a minimal cloud VM, but a single cheap VM would repeat EcoGoods current weaknesses. The estimate includes resilience, managed database protection, automated backup, security controls and global delivery, making it more appropriate for a growing international e-commerce business. Microsoft 365 Business Standard, if selected for office productivity, should be costed separately; Microsoft lists the annual subscription at USD 12.50 per user per month at the time of writing (Microsoft, 2026b).

4. Migration and Integration Strategy

EcoGoods should migrate through a phased, risk-controlled programme rather than a single big-bang cutover. The strategy should combine discovery, assessment, strategy selection, validation, transition and production monitoring with the 7Rs migration model. AWS defines the 7Rs as retire, retain, rehost, relocate, repurchase, replatform and refactor/re-architect, making it suitable for evaluating each EcoGoods workload separately (AWS, 2026a).

The first phase is assessment and planning. EcoGoods should inventory its Dockerised application, MySQL database, Docker Compose configuration, email server, file storage, HR/finance software, development workflows and manual backup process. Dependencies between the web application, database, product assets, email notifications and internal files must be mapped. The e-commerce platform and database should be prioritised because they directly affect sales, orders and customer experience.

The second phase is a proof of concept. A non-production AWS environment should be built using the proposed VPC, EKS, RDS, S3, IAM, WAF and monitoring controls. The Dockerised application should be deployed to EKS, while a test copy of MySQL is imported into RDS. The web application, database and files should be replatformed to EKS, RDS and S3; email and office tools should be repurchased as SaaS; the monthly SSD backup should be retired; and

difficult HR/finance systems may be retained temporarily with a review date. The workload-to-migration strategy mapping is summarised in Appendix Item D.

The third phase is data migration and synchronisation. A verified full backup should be taken before migration. AWS Database Migration Service can support full-load migration and change data capture, reducing the final cutover window (AWS, 2026e). Static files and product assets should move to S3 using AWS DataSync, which provides data verification options (AWS, 2026d). Email migration should run in parallel briefly to reduce communication disruption.

The fourth phase is validation and cutover. EcoGoods should complete functional testing of browsing, checkout, inventory updates and customer accounts; performance testing against campaign-level traffic; and security validation of IAM, WAF, encryption, private subnets and backup access. DNS should then be switched through Route 53 to CloudFront and the load balancer. A rollback plan should keep the on-premises server as a temporary warm standby until the AWS environment is stable.

After cutover, CloudWatch, CloudTrail, backup reports and application logs should be actively reviewed. Backup restoration must be tested regularly because an untested backup is not reliable disaster recovery. Capital One's AWS migration demonstrates that resilience improves only when cloud migration is supported by structured planning, monitoring and recovery testing (AWS, 2026g). For EcoGoods, success depends not only on moving workloads, but on proving that the new platform can operate, recover and scale reliably.

5. Conclusion

EcoGoods' current infrastructure is no longer proportionate to its scale, customer reach or operational risk profile. A single physical server may have been sufficient during the company's early stage, but it now creates serious constraints for a global e-commerce business handling customer traffic, order processing, inventory data, email services and internal files. Slow load times and downtime show that the present model lacks elasticity, while the monthly manual SSD backup creates significant data-loss and recovery risk.

The proposed AWS-based mixed cloud architecture provides a stronger foundation for future growth. EKS preserves EcoGoods' Docker-based deployment approach while adding orchestration, scalability and controlled releases. RDS for MySQL improves database

availability and recovery, while S3 and AWS Backup replace weak local backup with automated, durable and policy-driven protection. CloudFront, Route 53, load balancing and autoscaling improve customer experience across North America, Europe and Asia by reducing latency and increasing capacity during traffic spikes. SaaS adoption for email, office, HR and finance reduces non-core infrastructure management.

This is intentionally not a simple lift-and-shift migration. It combines replatforming, repurchasing, retaining and retiring so that each workload is treated according to business value, risk and technical need. With IAM, MFA, WAF, KMS encryption, private subnets, monitoring and restoration testing, the proposal improves scalability, availability, disaster recovery, operational efficiency and data protection. Overall, the cloud transition would enable EcoGoods to meet customer demand more reliably, protect business continuity and support continued international growth.

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Appendices

Appendix Item A - Cloud Service Model Mapping for EcoGoods

EcoGoods workload	Recommended model	Proposed service example	Reason
E-commerce application	PaaS / managed container service	Amazon EKS	Preserves Docker approach while adding orchestration, scaling and resilience.
Web traffic handling	Cloud edge/networking	Route 53, CloudFront, ALB	Improves global performance and distributes traffic away from a single server.
MySQL database	Database as a Service	Amazon RDS for MySQL Multi-AZ	Improves availability, failover, backup and recovery.
Files and product images	Storage as a Service	Amazon S3	Provides scalable and durable object storage.
Email and office tools	SaaS	Microsoft 365 or Google Workspace	Removes need to maintain an on-premises email and desktop-only office environment.
Backup and recovery	Managed backup / DR	AWS Backup, S3 lifecycle policies	Replaces manual monthly SSD backup with policy-based data protection.
Security and governance	Cloud security services	IAM, KMS, WAF, security groups	Protects identity, encryption, network access and web-facing services.

Source: Author's own work, based on the EcoGoods scenario and cloud service model analysis in Sections 2.1–2.4.

Appendix Item B - Proposed AWS Cloud Architecture for EcoGoods

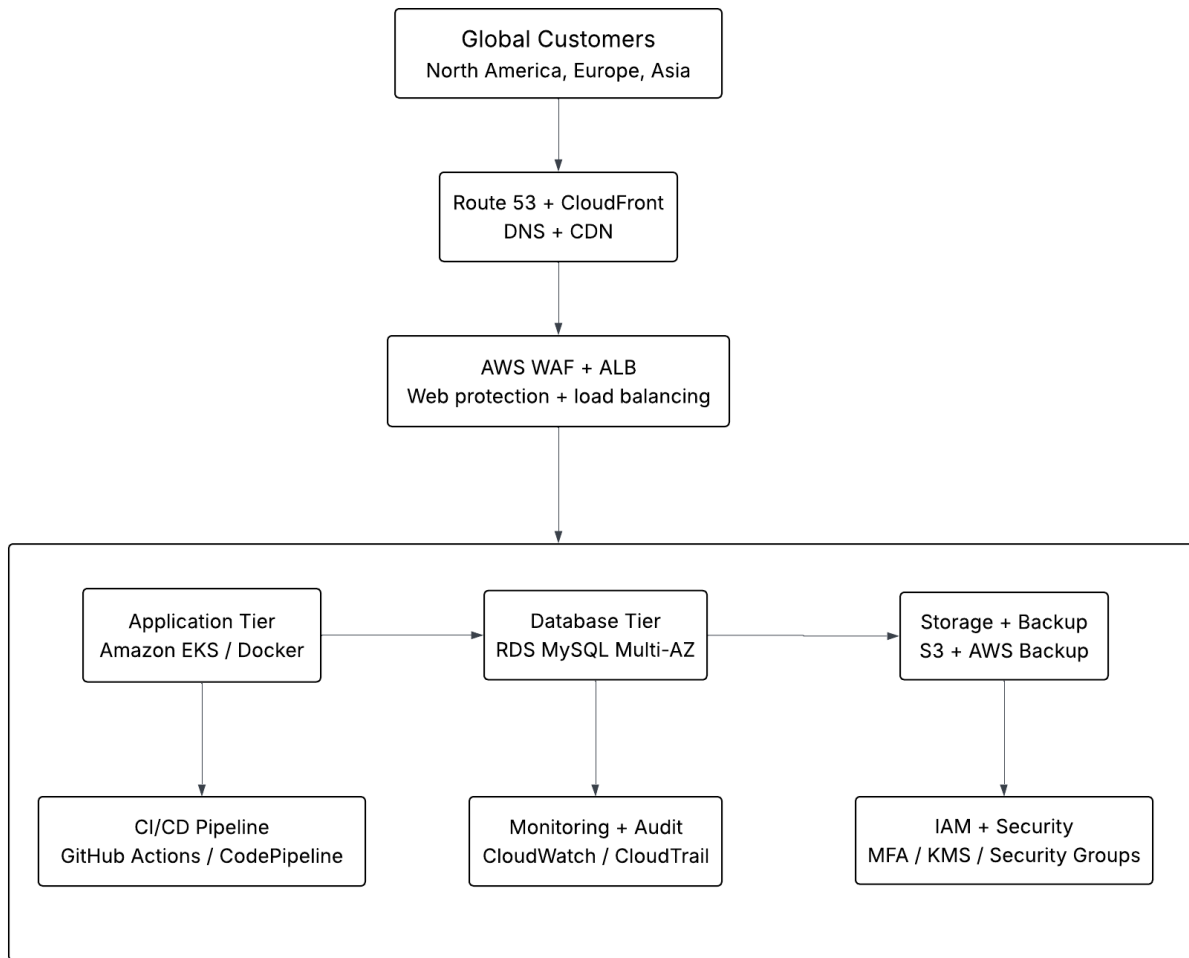


Figure B1: Proposed AWS cloud architecture for EcoGoods.

Source: Author's own work, based on the proposed AWS architecture discussed in Sections 2.2–2.4.

Appendix Item C - Cloud Security and Governance Controls

Risk	Recommended control	Purpose
Public database exposure	RDS in private database subnets with restrictive security groups	Prevents direct internet access to customer/order data.
Unauthorised access	IAM least privilege and MFA	Reduces account compromise and limits excessive permissions.
Web attacks	AWS WAF and HTTPS/TLS	Filters common application-layer attacks and protects customer sessions.
Data exposure	KMS encryption at rest and TLS in transit	Protects sensitive data and backups.
Accidental deletion or overwrite	S3 versioning and backup retention policies	Improves recovery from user error and corruption.
Poor visibility	CloudWatch monitoring and CloudTrail audit logging	Provides operational alerts and auditability.
Cost overrun	AWS Budgets, tagging and Cost Explorer	Controls usage-based cloud spending.

Source: Author's own work, based on the proposed AWS security controls, shared responsibility model and governance analysis in Section 2.4.

Appendix Item D - Migration Strategy Mapping

Current EcoGoods system	Migration approach	Target solution	Reason
Dockerised e-commerce app	Replatform	Amazon EKS	Keeps application logic while adding orchestration and scaling.
MySQL database	Replatform	Amazon RDS for MySQL	Retains MySQL compatibility while adding managed backup and failover.
Email server	Repurchase	Microsoft 365 or Google Workspace	Commodity service better delivered as SaaS.
File storage	Replatform	Amazon S3	Moves local files/assets to durable cloud object storage.
HR/finance desktop tools	Repurchase or retain temporarily	SaaS HR/finance tools or staged retention	Avoids rushed migration of sensitive business processes.
Manual SSD backup	Retire	AWS Backup and S3	Removes weak manual backup practice.
Existing on-premises server	Retain temporarily, then decommission	Warm standby during cutover	Supports rollback until the cloud platform is proven stable.
Full server lift-and-shift	Rehost	Not recommended as final state	Moving the server unchanged to a cloud VM would preserve many current weaknesses, including manual administration and limited resilience.
Existing virtualisation platform	Relocate	Not applicable	EcoGoods is Docker-based and there is no stated VMware or similar virtualisation estate to relocate.
Future application modernisation	Refactor / re-architect	Later phase after stable migration	Full refactoring is not recommended during initial migration, but CI/CD and EKS create a foundation

			for later microservices or serverless modernisation.
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Source: Author's own work, based on the migration and integration strategy discussed in Section 4.